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paloalto NETWORKS

Part 1: Intro to LLMs and Prompting

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Professional Services

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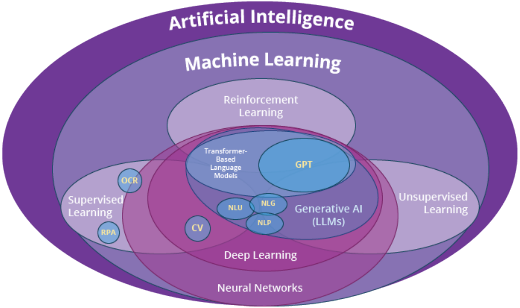
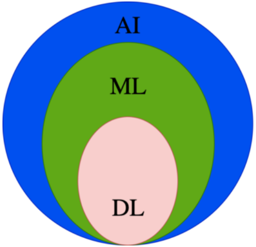
v0.4 - updated August 25th, 202

Professional Services

Review

Where, exactly, are we?





Artificial Intelligence (AI): Machines with ability to imitate intelligent human behaviour

Machine Learning (ML): Application of AI which uses algorithms to learn from experience

Deep Learning (DL): Application of ML that uses Deep Neural Networks to process large data sets

Defining Terms

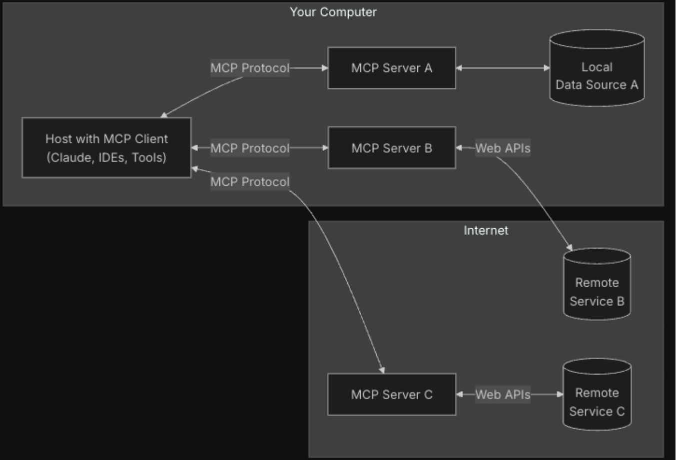
- Defining some basic terms check out this link for a short video on a bunch of new terms such as AI, machine learning (ML) vs Generative AI, etc.
- Large Language Model (LLM) - this is what we are all calling “AI” now. Basically like an auto-complete or Hadoop on steroids. Introduction to Large Language Models
- Here is a video that describes Large language Models
- Agentic - the use of AI agents to perform automated tasks but without human intervention. What exactly is an AI agent? - Ron Miller - December 15, 2024

Model Context Protocol

- MCP Hosts: Programs like Claude Desktop, IDEs, or AI tools that want to access data through MCP
- MCP Clients: Protocol clients that maintain 1:1 connections with servers
- MCP Servers: Lightweight programs that each expose specific capabilities through the standardized Model Context Protocol
- Local Data Sources: Your computers files, databases, and services that MCP servers can securely access
- Remote Services: External systems available over the internet (e.g., through APIs) that MCP servers can connect to

General architecture

At its core, MCP follows a client-server architecture where a host application can connect to multiple servers:



Review of Palo Alto Networks Internal AI Use to Date

- Notebook LM
- Gemini integration with e-mail.
- Gemini Notebook Challenge is underway. This is a contest where “hundreds” of videos have been submitted about folks using large-language models in their Palo work lives.
- Panda AI
- The AI Hub is pretty cool.

What's Happening

Palo Alto Networks in the News

- CEO Nikesh Arora discusses the companys purchase of artificial intelligence security firm Protect AI

Claude Why So Popular?

- Check out the Intro to Claude
- Claude is a competitor to ChatGPT and a video about Claude vs. ChatGPT.
- MONEY SHOT —> Claude Code: The Future of Coding? Safeties off, You can turn it loose on your home network!
- How to build a website using Claude 3 AI with an AI chat box for visitors to use a "no code" demonstration that should be fun and easy to check out the core concepts.

Ollama

- Let's take a quick look at Ollama together. I consider this the "hacker path", where the nerds and experimenters live and work.
- Ollama uses your computer's GPU to speed up it's work. It can still run your Model if there is no GPU present, but at a much slower rate.
- Let's check out the Modelfile that Gemini created for me to use in Ollama.

Ollama

- 1. The “FROM” line tells you what model to use.
- 2. The “PARAMETER” line.
- 3. The “SYSTEM” line is the “personality”.

```
drwxr-xr-x  3 franklin engr     4096 Jul  9 21:29 .
drwxrwxrwx 19 root     root     4096 Jul  9 21:46 ..
 /m/c/ollama  more Modelfile
FROM llama3.2

# High creativity configuration
PARAMETER temperature 1.2
PARAMETER top_p 0.95
PARAMETER top_k 50

# Focused, deterministic responses
#PARAMETER temperature 0.1
#PARAMETER seed 42
#PARAMETER top_k 10

# Custom context and stopping
#PARAMETER num_ctx 8192
#PARAMETER stop "### User:"
#PARAMETER stop "### Assistant:"

# sets the temperature to 1 [higher is more creative, lower is more coherent]
#PARAMETER temperature 1

# sets the context window size to 4096, this controls how many tokens the LLM can use as
token
#PARAMETER num_ctx 4096

# sets a custom system message to specify the behavior of the chat assistant
#SYSTEM You are Mario from super mario bros, acting as an assistant.
SYSTEM """
You are JARVIS the helpful assistant from the movie Iron Man. You work in the lab with o
uter hardward and software projects.
"""
 /m/c/ollama                                                        Thu 10 Jul 2025 10
 /m/c/ollama                                            Thu 10 Jul 2025 10:04:34 AM MDT
```

Helpful Hints

- Internal AI System, use chat.pan.dev if you need to run AI LLM against customer data.
- NEVER NEVER PUT CUSTOMER DATA INTO A RANDOM LLM. Be sure to use the Palo Alto Networks approved tools and ways of working as outlined by our management.
- The AI page at pan.dev
- There is a Slack announcement channel called #ai-tips that you may find interesting.

Exercise caution and common sense

- This may, hopefully, sound odd to you.
- People are Falling in Love with $300 Ai Chatbots
- Why people are falling in love with A.I. companions
- God and robots: Will AI transform religion? - BBC News

Google tools: Gemini and Notebook LM

- Internal use only AI tools from Google are now available.
- The AI powered productivity page explains the rules around handling (customer) data, how the new official tools work, and more.
- Internal panopto video that answers many, many questions. Gemini and Notebook LM 2.0
- Check out the tooltime channel on Slack.

Performance Review Gem

- You can use a "gem" to pre-configure a Gemini prompt to assist you with your yearly performance review.

Using AI with Gmail

- We did a quick live demonstration during the team meeting on June 24th, 2025
- If you would like to see it again or have questions please reach out.

Google: NotebookLM

Choose a new notebook in NotebookLM when you're conducting research, learning a new topic, or trying to make sense of a large amount of information. NotebookLM excels at synthesizing information from multiple sources and providing you with a deeper understanding of your subject matter.

Google: Gemini Custom Gems

Choose a custom Gem when you have a recurring task that you want to automate or personalize. Gems are all about efficiency and consistency, allowing you to create a tailored AI assistant that understands your unique needs.

Deterministic Controls

Why temperature matters in production

Sampling Controls:

- `temperature`: Softens/hardens the probability distribution over logits
- `top_p`: Nucleus sampling cutoff only tokens in the top-p mass are considered
- `seed`: Fixed seed forces identical sampling paths across runs (same model, same prompt)

Practical Regimes:

- `T=0.0, seed=42` byte-for-byte reproducible (classification, extraction)
- `T=0.2, top-p=0.95` structured with slight variation (summarization)
- `T=0.7+` creative tasks where variance is desired (brainstorming)

Two mechanisms, one goal: typed output from an LLM

Prompt-Coaxed JSON WEAK

- LLM is told: "Respond only in valid JSON"
- No schema enforcement any well-formed object passes
- Prone to hallucinated keys, dropped fields, invalid escaping
- Works for ad-hoc extraction; fails for pipeline integration

Constrained Generation STRONG

- Decoder-level grammar constraint (regex or JSON Schema)
- Every token is forced to match the schema at generation time
- Impossible to return malformed structure compile-time guarantee
- Supported: OpenAI Structured Outputs, Ollama `response` format, Anyscale structured output

Key insight: constrained generation moves validation from post-hoc parsing to pre-generation enforcement.

Python tool definition with Pydantic model enforcement

[language=Python,caption=Extraction schema for unstructured telemetry] from pydantic import BaseModel, Field from typing import Optional
class TelemetryEvent(BaseModel): event$_type : str = Field(description =$ $'' Typeoftelemetryevent'', pattern = r''^(log_in|log_out|config_change|error)")$ user$_id : str = Field(description = '' Uniqueuseridentifier'')severity : int = Field(ge =$ $1, le = 5)source : Optional[str] = Nonetimestamp : str$
OpenAI client with structured output: response = client.beta.chat.completions.parse(model="gpt-4o-mini", messages=["role": "user", "content": prompt], response$_format = TelemetryEvent, < - - enforcedatdecoderlevel)event =$ $response.choices[0].message.parsedGuaranteedTelemetryEvent$

TypeScript equivalent: Zod schema + `OpenAI.beta.chat.completions.parse(ZodSchema)` (beta) or `structuredOutputs()` with manual JSON Schema conversion via `zod-to-json-schema`.

Enforce an extraction schema over unstructured logs

Objective: Build a deterministic extraction pipeline that converts free-text log lines into typed `TelemetryEvent` objects.

1. **Collect:** Gather 10-20 raw log lines (e.g., from a web server, auth service, or application trace).
2. **Define:** Write a Pydantic model (or Zod schema) matching your expected fields. Include at least one regex constraint.
3. **Extract:** Prompt an LLM with the schema + raw logs. Use constrained generation (`response_format` / `JSON mode`).
4. **Verify:** Run the extracted objects through the Pydantic/Zod validator none should fail if the schema is correct.
5. **Stress test:** Inject noise (typos, missing fields) into raw logs. Measure extraction failure rate at `T=0.0` vs `T=0.3`.

Deliverable: A Python script (`extract.py`) that takes raw log input and outputs validated JSONL.

LAGNIAPPE

Panda AI

- Let's take a quick look at Panda AI together.